
When Phase Cancellation Meets Hardware Noise: Measurement Bottlenecks in Adaptive Berry Phases

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Abstract

Adaptive variational circuits compress cyclic many-body evolution, but their measured overlap is only one part of a Berry-phase estimator. We study the complete reconstruction, including noisy energy and geometric-connection measurements on a frozen variational trajectory. Independently implemented imaginary- and real-time adaptive simulations reproduce the quantized phases of four-site Su–Schrieffer–Heeger–Hubbard chains at interactions $U = 0$ and 10. Density-matrix simulations include entangling-gate depolarization, readout errors, relaxation, idle dephasing, and independent finite sampling of every phase contribution. At 10^6 shots per observable, a predeclared 0.1-radian, 95th-percentile accuracy criterion gives conservative two-phase depolarizing thresholds of 2.17×10^{-4} and 3.23×10^{-6} for the two interactions. These are thresholds along a specified joint device profile, not independent gate-error limits. The principal failure is a biased phase integral: unequal circuit histories spoil forward–return cancellation even while endpoint interference remains visible. Independent shot fluctuations also add across the three phase contributions. Idealized noise-rate extrapolation improves the thresholds by factors 4.7 and 13.6 at the same total shot budget. The resulting distinction between state fidelity, overlap visibility, and integrated-phase accuracy identifies a measurement bottleneck that circuit depth alone misses.

1 Geometric protection does not protect every estimator

A topological invariant is a property of a state family; its experimental reconstruction is a statistical procedure. These two forms of robustness need not coincide. Berry phases underpin polarization and topological classification [Berry, 1984, Xiao et al., 2010], and quantum algorithms can obtain them through cyclic evolution or overlaps between variational ground states [Murta et al., 2020, Tamiya et al., 2021, Xiao et al., 2023]. Adaptive variational quantum dynamics simulation (AVQDS) offers an attractive way to compress long evolutions into short, problem-dependent circuits [Yao et al., 2021, Zhang et al., 2025, Mootz and Yao, 2025]. The remaining question is whether the complete phase reconstruction inherits that compression’s apparent noise advantage.

In AVQDS, global phase is absent from the density-matrix variational principle. An endpoint Hadamard test supplies an overlap phase φ_{qe} , while device measurements integrated along the trajectory supply a phase correction φ_G . Their sum is the observable Berry-phase estimate. Accurate noiseless cancellation therefore does not imply that measuring only the endpoint diagnoses the method’s resilience. An energy bias is multiplied by an evolution-time integral; connection measurements have their own circuits and sampling errors. Moreover, the two halves of an adaptively constructed loop generally execute circuits of different sizes.

We characterize this distinction on an eight-qubit interacting model with a ninth readout ancilla. The study separates the three phase errors, measures their covariance, compares channel-isolated errors, resolves a finite-shots axis, and tests an explicitly idealized mitigation procedure. The numerical question is deliberately local: how much execution noise can a fixed, independently grown trajectory

tolerate? Relearning the trajectory from noisy variational equations is outside this experiment. Neither the small system size nor the thresholds below establish a scalable quantum advantage.

2 A reproducible phase experiment

Model and independent reference. With sites $j = 0, \dots, 3$, periodic boundaries, hopping scale $t = 1$, and spin σ , we use

$$H(\lambda) = \sum_{j,\sigma} \left[t_j c_{j+1,\sigma}^\dagger c_{j,\sigma} + \text{h.c.} \right] + U \sum_j n_{j\uparrow} n_{j\downarrow} - \frac{U}{2} \sum_{j,\sigma} n_{j\sigma}, \quad t_j = 1 + (-1)^j \delta. \quad (1)$$

The boundary hopping for spin up acquires $e^{-i\lambda}$; spin down remains untwisted. We construct the Jordan–Wigner expansion algebraically in interleaved site/spin order. Its nonidentity pool contains 18 strings for $U = 0$ and 22 for $U = 10$. All four experiments use $\delta = \pm 0.6$ and loop time $T = 20$ in inverse-hopping units. Independent diagonalization in the two-up, two-down sector gives a gauge-invariant Wilson product on 512 points; a 256-point calculation checks convergence. No reference eigenstate or phase enters adaptive growth or noisy estimation. The two signs of loop-orientation convention agree modulo 2π for these quantized references.

Adaptive preparation and propagation. The state is $|\psi(\theta)\rangle = U_P(\theta_P) \cdots U_1(\theta_1)|b\rangle$, with $U_\mu = e^{-i\theta_\mu A_\mu}$. AVQITE uses the 616 two- and four-qubit X/Y strings with odd Y parity. The computational reference has spin-up occupation on the first two sites and spin-down on the last two for $\delta < 0$. For $\delta > 0$, it is 11000011 in the stored bit convention, following the reference archive’s explicit preparation change. This distinction is necessary: the former reference has zero overlap with the noninteracting positive-dimerization ground state. Preparation uses an imaginary-time residual cutoff 10^{-3} , a variance target 10^{-8} , and a time cap of 60; the cap is reached for both interacting preparations. Their attained errors are reported below.

AVQDS reoptimizes all preparation and newly inserted angles. Writing $\Pi_\perp = 1 - |\psi\rangle\langle\psi|$ and $D_\mu = \Pi_\perp \partial_\mu |\psi\rangle$, the equation solved is

$$\text{Re}(D^\dagger D)\dot{\theta} = \text{Re}(D^\dagger v), \quad L^2 = 2\|v - \sum_\mu D_\mu \dot{\theta}_\mu\|^2, \quad (2)$$

where $v = -i\sigma(u)(H - E)|\psi\rangle$ in real time and $v = -(H - E)|\psi\rangle$ in imaginary time. Here $u \in [0, T]$ is increasing loop progress, $\lambda = 2\pi u/T$, and $\sigma = +1$ before $T/2$ and -1 afterwards. This implements forward and reversed Hamiltonian evolution with a continuously increasing twist. A minimum-norm eigensolve discards eigenvalues below $\max(10^{-9}, 10^{-9}\lambda_{\max})$. Candidates are ranked by their reduction of L^2 ; each growth batch appends disjoint supports with zero angles, following the TETRIS construction [Zhang et al., 2025]. Growth enforces $L^2 < 10^{-4}$ at each accepted-step start. RK4 caps every stage’s angle change at 0.01, limits time steps to 0.04, and splits exactly at the reversal. Diagnostics against exact states are computed only after the trajectory is complete.

Phase and sign conventions. For the physical ket $e^{ig}|\psi(\theta)\rangle$, differentiating the Schrödinger equation fixes

$$\varphi_B = \arg\langle\psi(0)|\psi(T)\rangle + G_1 + G_2 \pmod{2\pi}, \quad (3)$$

$$G_1 = -\int_0^T \sigma(u)E(u) du, \quad G_2 = -\int_0^T \text{Im}\langle\psi|\dot{\psi}\rangle du. \quad (4)$$

This sign is tied to the declared $e^{-i\theta A}$ ansatz and bra–ket ordering. Some formulas in the supplied manuscript print a plus sign before the latter imaginary part. We explicitly audit that alternative: its errors are listed in the computational record and it fails our direct-propagation check. Equation (4) agrees with independently integrated Schrödinger evolution. Thus our convention correction is disclosed and tested, rather than treated as an unrecorded phase adjustment.

The endpoint circuit prepares $|\psi(0)\rangle$ and $|\psi(T)\rangle$ on the two ancilla branches. Separate X and Y quadratures give $\hat{\varphi}_{\text{qc}} = \text{atan2}(\hat{y}, \hat{x})$. This avoids the zero derivative of an inverse-cosine estimator at quantized phases, although it still fails when interference visibility vanishes. Integrals are accumulated without folding. All errors use $d(\alpha, \beta) = |\arg e^{i(\alpha-\beta)}|$; displayed means are aligned to the ED reference. Classification chooses the closer of the two ED phase classes on the circle.

Circuits for the correction. For a Pauli-product ansatz the connection has the exact reduction

$$-\text{Im}\langle\psi|\partial_\mu\psi\rangle = \langle\psi_{<\mu}|A_\mu|\psi_{<\mu}\rangle, \quad |\psi_{<\mu}\rangle = U_{\mu-1}\cdots U_1|b\rangle. \quad (5)$$

We cancel the common suffix algebraically before compilation and directly measure this Pauli string on the remaining prefix. Energy terms use the full state-preparation circuit at that node. This is a specified measurement compilation of the same connection, not a noise-free connection oracle. Both types receive the identical physical channels below. Each Pauli rotation is compiled into basis rotations, a CX parity network, an R_z , and its inverse network; a controlled rotation uses two additional CXs. We retain active zero-angle rotations. Computational product-state initialization is ideal. Output-state infidelity is a classical density-matrix diagnostic of a separately executed, unconditional eight-qubit output-preparation circuit against the ED ground state; it is not estimated from device shots or used by the phase estimator.

A cubic Hermite interpolant of the accepted trajectory supplies panel midpoints and angle increments. We integrate G_2 by midpoint connection times the angle increment, and G_1 by signed midpoint energy times panel width. Starting with 64 panels, we double until each leg differs from the full RK4 integral by less than 0.0025 radians. The resulting node counts are 2048, 512, 1024, 1024. The midpoint reversal is an exact boundary. Newly inserted generators with nonzero change within a panel are included. The initial refinement used noiseless data. A subsequent noisy resolution check exposed discontinuities in energy bias at ansatz-growth events for $U = 0, \delta > 0$; we promoted that case from 128 to 512 nodes and repeated its full sweep. The original pilot is retained, and the threshold rule is unchanged. All rejected grids and a further near-threshold resolution check are recorded.

Joint device profile. Our all-to-all, serialized CX model is inspired by public trapped-ion timings, including the IonQ Aria profile [Microsoft Azure Quantum, 2026], but is not a simulation of its native compiler or a vendor calibration. At noise scale s , every CX on pair a, b is followed by

$$\mathcal{D}_{p_2}(\rho) = (1 - p_2)\rho + \frac{p_2}{4}I_{ab} \otimes \text{Tr}_{ab}\rho, \quad p_2 = 0.004s. \quad (6)$$

Its average gate infidelity is $3p_2/4$, so depolarizing probability and infidelity must not be conflated. Every qubit, including idling qubits and the ancilla, undergoes zero-temperature amplitude damping and dephasing with $T_1 = 50/s$ seconds and $T_2 = 1/s$ seconds. A single-qubit gate takes $135\mu s$ and a CX takes $600\mu s$. Channels obey $\rho_{11} \mapsto e^{-\tau/T_1}\rho_{11}$ and $\rho_{01} \mapsto e^{-\tau/T_2}\rho_{01}$. The corresponding pure dephasing rate is $1/T_2 - 1/(2T_1)$. Symmetric independent readout flips have probability $r = 0.0039s$ on every measured qubit. Measured X/Y bases use one simultaneous rotation per qubit, with its relaxation included. There is no additional single-qubit depolarization, leakage, crosstalk, or coherent calibration error. At $s = 0$ coherence times are infinite. Thresholds co-vary all these rates in the stated ratio.

Density matrices have dimension 256 for correction circuits and 512 for endpoint interference. Commuting idle channels past gates on other qubits is exact and permits lazy application. Identical prefixes are shared within the classical simulation; on hardware each measurement is a separate preparation. Energy and connection estimates use actual independent binomial counts from their noisy parity probabilities. The two quadratures are independently sampled as well. Deterministic identity terms need no shots. The simulator averages the CPTP channel exactly; its finite measurement samples include the channel-induced change of outcome statistics.

Accuracy, sampling, and threshold extraction. The protocol was saved before the noise sweep. The reference budget is $S_{\text{ref}} = 10^6$ shots per observable, including each endpoint quadrature. We use 256 independent seeds, numbered 0–255, at every configuration and noise strength. Accuracy means that the 95th percentile of absolute circular error is at most $\epsilon = 0.1$ radians. On an increasing grid, we take the first crossing of this bound using linear interpolation of the cumulative maximum of the empirical 95th percentiles. This prevents phase wrapping at larger noise from creating an artificial recovery of the threshold. Midpoint refinements continue until the crossing bracket is at most 25% of its interpolated location. A 1,000-resample bootstrap independently resamples seeds at each strength and supplies a conditional 95% interval. We also report the sampled crossing bracket, which is distinct from sampling uncertainty. The smaller threshold within each interaction regime defines its conservative two-phase operating limit.

An independent shot sweep uses $S \in \{10^3, 3 \times 10^3, 10^4, 3 \times 10^4, 10^5, 3 \times 10^5, 10^6, 3 \times 10^6, 10^7\}$ at $s = 0$. Its floor is the smallest tested budget satisfying the same criterion, with the preceding failed

budget reported. The complete strength grids, aggregate errors, and seeds are included in the record and Table 4; no strength is discarded after evaluation.

Exploratory mitigation. After the raw sweep began, we separately registered first-order noise-rate extrapolation [Temme et al., 2017, Cai et al., 2023]. Every observable, including both quadratures, is estimated as $2\hat{m}(s) - \hat{m}(2s)$ before phase reconstruction. Each scale receives $S_{\text{ref}}/2$ shots, so the total shots per observable match the raw estimator. Counts at the two scales are independent and extrapolated values are not clipped. The same accuracy and first-crossing rule is used. This experiment assumes controllable scaling of all channel rates; it does not establish that physical gate folding produces that scaling on a particular device.

3 Independent anchors and a complete noise sweep

The independently computed ED phases distinguish negative-dimerization nontrivial states from positive-dimerization trivial states. Table 1 compares our actual trajectories to matching archive traces at $T = 20$. The maximum noiseless circular error is 2.16×10^{-4} radians. The interacting nontrivial endpoint is appreciably nonadiabatic, with infidelity 0.330, while its phase error remains 2.16×10^{-4} radians. Direct real-time propagation and the variational trajectory agree closely; this verifies the deterministic cancellation without assuming it from ground-state fidelity.

Resource agreement is qualitative, not exact. Our eigentruncated solves produce 10–12 Pauli layers in the noninteracting cases and 63–65 in the interacting cases. The supplied archive has larger counts and a different dimerization ordering; in particular, its positive-dimerization interacting trace is also costly. We do not reproduce a universal rule that the nontrivial case must cost more. A stricter $L^2 = 10^{-6}$ noninteracting control retains 88 CXs and reduces the phase error to 3.80×10^{-6} radians, showing that the compressed count does not come from relaxing the residual tolerance. These independently grown circuits, not the archive’s saved ansatz, define all noise thresholds here.

Table 2 reports the unrounded-trajectory anchor and a measured point immediately below each raw threshold. At 10^6 shots per observable, both topological labels are correctly distinguished below the conservative per- U limit in all 256 sampling repeats. The joint thresholds are far more stringent for $U = 10$. The equivalent device parameters, including readout and coherence times, are in Table 3. A threshold in p_2 is conditional on these other rates and on the serialized schedule; it is not a standalone gate specification.

The observed shot floors are 3,000, 3,000, 10,000, 10,000 shots per observable, ordered as $(U, \delta) = (0, -0.6), (0, +0.6), (10, -0.6), (10, +0.6)$. These are grid-resolved floors, not exact minima. Their seed-bootstrap intervals on the tested shot grid are [3, 000, 3, 000], [1, 000, 3, 000], [10, 000, 10, 000], [10, 000, 30, 000]. Figure 2 separates sampling uncertainty from the nonzero numerical integration bias. Independent measurements exhibit the expected inverse-square-root shot dependence over the resolved range; the fitted exponents are reported in the figure and logs. At high shots the total error approaches the quadrature and variational bias, so an error curve need not continue with the same slope indefinitely.

4 Why the phase integral fails first

Let δ_q , δ_1 , and δ_2 denote the signed errors in the endpoint phase and the two integrals, aligned continuously near the operating point. Locally,

$$\Delta\varphi = \delta_q + \delta_1 + \delta_2, \quad \text{Var}(\Delta\varphi) = \sum_{j=q,1,2} \text{Var}(\delta_j) + 2 \sum_{j < k} \text{Cov}(\delta_j, \delta_k). \quad (7)$$

The frozen trajectory correlates deterministic approximation errors; it does not correlate independent measurement outcomes. Our three sampling streams are independent by construction, and the measured covariance test in Table 5 agrees with variance addition within finite-seed fluctuations. The relevant finite-shots scaling for the integrals is

$$\text{Var}(\hat{G}_1) = \sum_{k,a} w_{ka}^2 \frac{1 - m_{ka}^2}{S}, \quad \text{Var}(\hat{G}_2) = \sum_{k,\mu} (\Delta\theta_{k\mu})^2 \frac{1 - c_{k\mu}^2}{S}, \quad (8)$$

Table 1: Noiseless anchors and matched archive comparison. The phase columns give circular errors against independently computed ED. Preparation infidelity is against ED; dynamics infidelity is the maximum sampled error against independent real-time propagation. Resource entries are Pauli layers/CXs including preparation. Archive values are verification references, not our measurements. N/T denotes nontrivial/trivial in figures.

U, δ	Our phase err.	Archive err.	Prep. infid.	Dynamics err.	Our resources	Archive
0, -0.6	1.90×10^{-4}	2.83×10^{-4}	8.50×10^{-10}	4.65×10^{-6}	10/88	44/464
0, +0.6	3.51×10^{-5}	8.33×10^{-5}	8.50×10^{-10}	8.64×10^{-6}	12/112	20/212
10, -0.6	2.16×10^{-4}	4.95×10^{-4}	3.68×10^{-6}	3.05×10^{-6}	65/646	86/872
10, +0.6	1.46×10^{-5}	2.40×10^{-3}	1.62×10^{-6}	6.32×10^{-6}	63/628	103/1100

Table 2: Primary results at 10^6 shots per observable. Measured rows are the largest sampled strengths below the first threshold. Phase is the circular mean aligned to ED, with one seed standard deviation. Error is exact circular error for noiseless rows and its empirical 95th percentile for sampled rows. Threshold brackets are bootstrap 95% intervals for depolarizing probability p_2^* along the joint profile. D_P counts Pauli layers; CXs count the output-state circuit.

U, δ	Execution	Berry phase (rad)	Error (rad)	p_2^* [95% CI]	D_P	CX
0, -0.6	noiseless	3.14140	1.90×10^{-4}	$(3.58 [3.56, 3.59]) \times 10^{-4}$	10	88
0, -0.6	$s = 0.085$	3.23330 ± 0.00218	0.0955	$(3.58 [3.56, 3.59]) \times 10^{-4}$	10	88
0, +0.6	noiseless	0.00004	3.51×10^{-5}	$(2.17 [2.15, 2.18]) \times 10^{-4}$	12	112
0, +0.6	$s = 0.05$	0.08955 ± 0.00201	0.0929	$(2.17 [2.15, 2.18]) \times 10^{-4}$	12	112
10, -0.6	noiseless	3.14138	2.16×10^{-4}	$(3.82 [3.79, 3.86]) \times 10^{-6}$	65	646
10, -0.6	$s = 0.000875$	3.22760 ± 0.00364	0.0925	$(3.82 [3.79, 3.86]) \times 10^{-6}$	65	646
10, +0.6	noiseless	0.00001	1.46×10^{-5}	$(3.23 [3.20, 3.25]) \times 10^{-6}$	63	628
10, +0.6	$s = 0.00075$	0.08589 ± 0.00461	0.0931	$(3.23 [3.20, 3.25]) \times 10^{-6}$	63	628

where m and c are noisy measured expectations, and w includes the signed energy quadrature and Hamiltonian coefficient. Equation (8) explains the observed scaling; all reported outcomes come from binomial draws, not this variance formula.

The dominant bias has a different origin from finite- T cancellation. Writing $E_{\text{noisy}} = E + \Delta E$, the energy leg acquires

$$\Delta G_1 = - \int_0^{T/2} \Delta E(u) du + \int_{T/2}^T \Delta E(u) du. \quad (9)$$

Equal mirrored energy biases would cancel. Adaptive growth and the corresponding gate/idle histories break this equality. Depolarization also pushes interaction correlators toward zero, while amplitude damping changes occupations. For $U = 10$, these energy changes are weighted by the interaction scale and integrated over the two half loops. A small error per physical operation can therefore generate a large signed phase drift. Connection bias contributes separately through the distribution of angle increments and the lengths of its preparation prefixes. It need not have the sign that cancels the energy drift.

The diagnostics resolve three distinctions. First, endpoint visibility remains 0.811–0.994 at the raw operating limits: loss of an observable endpoint fringe is not what fixes those limits. Second, the full-method error can grow while the endpoint-only phase error stays small; the latter is a narrower diagnostic and is never used as our reported threshold. Third, circuit count alone is insufficient. The interacting circuits are longer, but interaction-weighted energy bias and forward/return exposure amplify their disadvantage beyond their CX-count ratio. At these diagnostic strengths, entangling-gate depolarization dominates the noninteracting bias, while relaxation/dephasing is slightly larger than depolarization in the interacting cases. Symmetric readout errors contribute much less in this profile. The channel-isolation results quantify these contributions without assigning an independent threshold to a single channel. We also test a compiler intervention using exact noisy channel means: every energy circuit is padded with the complete final generator list, with future angles zero, while connection and endpoint circuits remain unchanged. This removes gate-count variation without changing any ideal state. In the interacting cases it retains 91.9% and 89.2% of the original energy-leg bias. Thus count asymmetry alone explains only a small part of their drift: gate angles, ordering, and

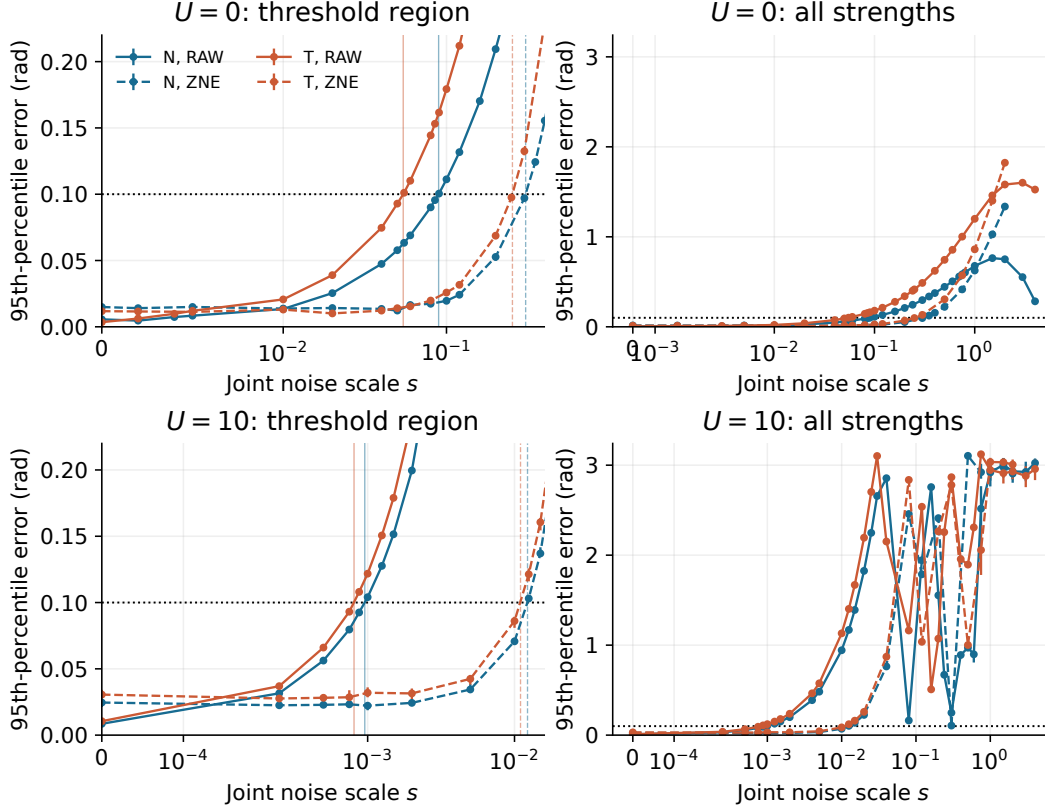


Figure 1: Full-method phase error versus the joint noise scale. The left pair resolves the first threshold crossing; symbols show empirical 95th-percentile circular errors and bootstrap intervals, with the predeclared 0.1-radian boundary dashed. Raw and extrapolated estimators use the same total shots per observable. The right panels show the full explored range: nonlinear drift and phase wrapping make high-noise errors nonmonotone. Every sampled strength is represented; first-crossing extraction excludes later apparent recoveries.

Table 3: Device interpretation and equal-shot extrapolation. Coherence times at the raw threshold are in seconds. The grid bracket describes numerical resolution separately from bootstrap uncertainty. ZNE uses 5×10^5 shots at each of s and $2s$; its quoted p_2 is the lower physical rate. The depolarizing contribution to average CX infidelity is $3p_2/4$.

U, δ	Raw s^*	Grid bracket	Readout r^*	T_1/T_2 (s)	ZNE p_2^* [95% CI]
0, -0.6	0.089513	[0.085, 0.09]	3.49×10^{-4}	559/11.2	$(1.22 [1.21, 1.23]) \times 10^{-3}$
0, +0.6	0.054331	[0.05, 0.055]	2.12×10^{-4}	920/18.4	$(1.01 [1.00, 1.02]) \times 10^{-3}$
10, -0.6	0.00095608	[0.000875, 0.001]	3.73×10^{-6}	52297/1045.9	$(4.90 [4.88, 4.97]) \times 10^{-5}$
10, +0.6	0.00080782	[0.00075, 0.000875]	3.15×10^{-6}	61895/1237.9	$(4.39 [4.31, 4.51]) \times 10^{-5}$

the associated noisy preparation histories also matter. This control is a mean-bias diagnostic, not an additional finite-shot threshold.

Rate extrapolation cancels a first-order expectation bias: if $m(s) = m(0) + as + bs^2 + \dots$, then $2m(s) - m(2s) = m(0) - 2bs^2 + \dots$. It also raises variance. With equal total shots and comparable single-shot variances, the factor is ten, because the two coefficients are 2 and -1 and each receives half the shots. Our sampled results include this cost. The improved thresholds in Table 3 are 1.22×10^{-3} , 1.01×10^{-3} , 4.90×10^{-5} , 4.39×10^{-5} in configuration order. The mitigation changes estimates, not the noisy output-state fidelities. Its gains thus support the phase-bias diagnosis under the stated noise-scaling assumption.

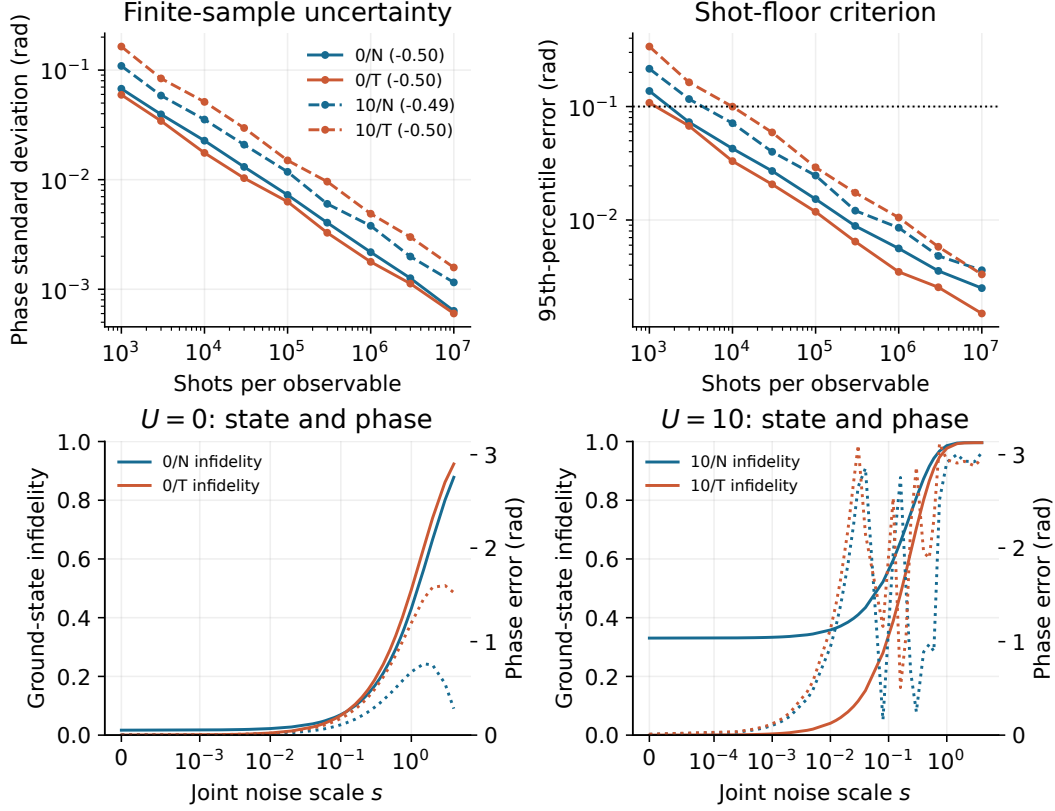


Figure 2: Distinct accuracy axes. Top: observed phase standard deviation and 95th-percentile error versus shots per observable at zero incoherent noise; fitted slopes use finite-binomial samples. Bottom: output-state infidelity against the ED ground state (solid) and the corresponding phase error (dotted). The nontrivial interacting trajectory begins with finite diabatic infidelity, and phase accuracy is lost well before the noisy endpoint state becomes maximally mixed.

5 Failure modes and the cost of resolving them

Three numerical failures were found and corrected before the final analysis. An unsuitable product reference converged to an excited state in the noninteracting trivial regime; changing to the archive-documented product reference restored ground preparation without consulting an ED state during growth. A literal global-phase sign from the supplied text failed an independent propagation identity. Finally, a coarse 64-node connection quadrature produced an error as large as 1.331 radians, although its energy integral was already accurate. Variational coordinates can move rapidly in nearly redundant directions, so a smooth physical state does not guarantee a smooth coordinate-level phase integrand. The full convergence sequence and a finer-grid noise check are retained; the latter changes near-threshold bias by at most 2.99×10^{-3} radians.

At higher physical noise the endpoint visibility eventually approaches the sampling scale, making the angle estimate unstable. Separately, large integral drifts can wind through multiples of 2π . A recovered small circular error at an isolated high-noise point is therefore not evidence of a safe operating range. The cumulative first-crossing rule prevents this aliasing from inflating the reported threshold. The full-range curves expose these failures rather than suppressing them.

The most demanding output circuit has 1214 CXs, but endpoint depth substantially understates the total experiment. At the chosen quadratures, the number of independently sampled observables per phase estimate ranges from 18,586 to 154,652. At S_{ref} this requires 1.86×10^{10} – 1.55×10^{11} shots. Table 6 includes an ideal serialized lower-bound runtime obtained by summing actual compiled preparation durations weighted by shot count; measurement, reset, transport, classical control, and calibration overheads are excluded. A classical shared-prefix simulation can reuse a density matrix;

Table 4: Complete noise-strength inventory for the final primary sweeps. Each entry has seeds 0–255. Figures report their aggregate errors; the JSONL records retain every phase and error. Additional channel controls use $s = 0.05$ for $U = 0$ and $s = 0.00075$ for $U = 10$, with the same seeds. Superseded quadrature pilots and unsuccessful initialization are retained and identified in the attempt log.

U, δ	Method	Executed joint noise scales s
0, -0.6	RAW	0, 0.002, 0.004, 0.005, 0.01, 0.02, 0.04, 0.05, 0.055, 0.06, 0.08, 0.085, 0.09, 0.1, 0.12, 0.16, 0.2, 0.24, 0.3, 0.35, 0.4, 0.5, 0.6, 0.7, 0.75, 0.8, 1, 1.5, 2, 3, 4
0, -0.6	ZNE	0, 0.002, 0.005, 0.01, 0.02, 0.04, 0.05, 0.06, 0.08, 0.1, 0.12, 0.2, 0.3, 0.35, 0.4, 0.5, 0.75, 1, 1.5, 2
0, $+0.6$	RAW	0, 0.002, 0.004, 0.005, 0.01, 0.02, 0.04, 0.05, 0.055, 0.06, 0.08, 0.085, 0.09, 0.1, 0.12, 0.16, 0.2, 0.24, 0.25, 0.3, 0.4, 0.5, 0.6, 0.75, 1, 1.5, 2, 3, 4
0, $+0.6$	ZNE	0, 0.002, 0.005, 0.01, 0.02, 0.04, 0.05, 0.06, 0.08, 0.1, 0.12, 0.2, 0.25, 0.3, 0.5, 0.75, 1, 1.5, 2
10, -0.6	RAW	0, 0.00025, 0.0005, 0.00075, 0.000875, 0.001, 0.00125, 0.0015, 0.002, 0.004, 0.005, 0.01, 0.0125, 0.015, 0.02, 0.025, 0.03, 0.04, 0.08, 0.12, 0.16, 0.2, 0.24, 0.3, 0.4, 0.5, 0.6, 0.75, 1, 1.5, 2, 3, 4
10, -0.6	ZNE	0, 0.00025, 0.0005, 0.00075, 0.001, 0.002, 0.005, 0.01, 0.0125, 0.015, 0.02, 0.04, 0.08, 0.12, 0.2, 0.3, 0.5, 0.75, 1, 1.5, 2
10, $+0.6$	RAW	0, 0.00025, 0.0005, 0.00075, 0.000875, 0.001, 0.00125, 0.0015, 0.002, 0.004, 0.005, 0.01, 0.0125, 0.015, 0.02, 0.025, 0.03, 0.04, 0.08, 0.12, 0.16, 0.2, 0.24, 0.3, 0.4, 0.5, 0.6, 0.75, 1, 1.5, 2, 3, 4
10, $+0.6$	ZNE	0, 0.00025, 0.0005, 0.00075, 0.001, 0.002, 0.005, 0.01, 0.0125, 0.015, 0.02, 0.04, 0.08, 0.12, 0.2, 0.3, 0.5, 0.75, 1, 1.5, 2

Table 5: Measured decomposition at diagnostic strengths. Biases and standard deviations are in radians. R_V is observed total variance divided by the sum of the three leg variances; deviations from one reflect finite-seed covariance fluctuations. The columns separate endpoint, energy, and connection errors.

U, δ	s	b_q	b_1	b_2	$\sigma_q/\sigma_1/\sigma_2$	R_V
0, -0.6	0.05	+0.00019	+0.06069	−0.00638	0.0008/0.0005/0.0020	1.037
0, $+0.6$	0.05	+0.00147	+0.11097	−0.02293	0.0011/0.0007/0.0015	0.981
10, -0.6	0.00075	+0.00001	+0.05068	+0.02340	0.0012/0.0020/0.0025	1.055
10, $+0.6$	0.00075	+0.00008	+0.06599	+0.01980	0.0010/0.0020/0.0041	0.998

a device cannot reuse a destructively measured prefix in this way. These execution costs exclude learning the variational trajectory and are already a substantial barrier when individual circuits are shallow.

The primary study contains 168,960 noisy integrand-node evaluations and 64,256 finite-sampling phase runs. The sum of logged per-strength simulation wall times is 0.56 hours; some jobs overlap, so this is not total elapsed time. The artifact records individual timings and coordinator memory. Each complex128 density matrix occupies 1 MiB at eight qubits and 4 MiB at nine; processes use one or a small number at a time. Density jobs use pools of 12–24 workers and sampling jobs use 4–16, with single-threaded BLAS. No GPU, precomputed ansatz, or ground-state oracle is used in the variational evolution. Density-matrix costs are exponential in system size, so this approach is a validation tool, not a proposed scalable simulator.

6 Scope and relation to prior work

Cyclic gate-based Berry estimation was introduced by [Murta et al. \[2020\]](#); variational overlap approaches provide a different route to geometric quantities [\[Tamiya et al., 2021, Xiao et al., 2023\]](#). The latter’s topological robustness does not directly establish the noise tolerance of a time-integrated global-phase correction. Our experiment implements adaptive imaginary-time preparation [\[Gomes et al., 2021\]](#), real-time McLachlan evolution [\[Yao et al., 2021\]](#), and disjoint-support growth [\[Zhang et al., 2025\]](#), applying the phase reconstruction of [Mootz and Yao \[2025\]](#). The new evidence concerns execution noise in every measured phase leg and its practical sampling cost. It complements the observation that geometric and dynamical gates can have equivalent noise sensitivity under appropriate conditions [\[Colmenar et al., 2022\]](#); geometric origin alone is not an error bound. Extrapolation and its overhead are established techniques [\[Temme et al., 2017, Cai et al., 2023\]](#); our contribution is their measured effect on this complete estimator, with an unchanged total shot budget.

The conclusions are conditional. Only four configurations of a four-site chain are studied, away from the gap-closing transition. The ansatz and angles are frozen, so noisy metric/force estimation,

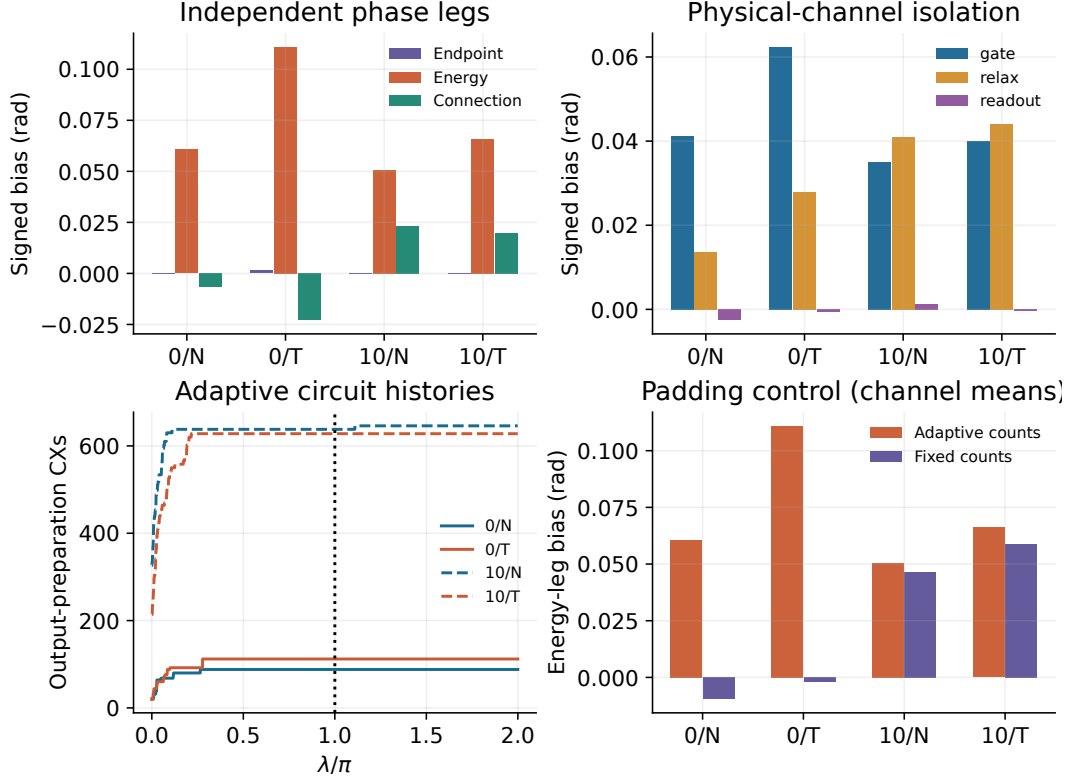


Figure 3: Origin of the error. Upper left: signed bias in each measured phase contribution at the diagnostic strengths. Upper right: phase bias with gate depolarization, relaxation/dephasing, or readout noise isolated at the same scale. Lower left: adaptive CX histories. Lower right: fixing the energy-circuit gate counts removes only a small fraction of the interacting energy bias. Channel-isolated simulations use the actual circuits; phase-component decompositions use independently sampled measurements.

Table 6: Measurement cost per phase estimate. Years are serialized gate-time lower bounds at the reference shots; reset, measurement, transport, and control overheads are excluded. S^* is the tested zero-noise shot floor per observable. Endpoint CXs include controlled preparation of both branches.

U, δ	Nodes	Observables	Total shots	Years	S^*	Endpoint CX
0, -0.6	2,048	70,718	7.07×10^{10}	104.1	3,000	154
0, +0.6	512	18,586	1.86×10^{10}	33.0	3,000	182
10, -0.6	1,024	154,328	1.54×10^{11}	1393.9	10,000	1214
10, +0.6	1,024	154,652	1.55×10^{11}	1337.7	10,000	1116

unstable feedback, and noise-dependent growth are excluded. Interacting AVQITE reaches a time cap with small but nonzero preparation error. Resource counts differ from the reference archive because independent growth and numerical regularization produce different circuits. Serialized gates, ideal product initialization, Markovian homogeneous channels, and exact rate control for mitigation are explicit modeling choices. No hardware experiment, Trotter noise comparison, routing optimization, or device-specific performance guarantee is claimed. Bootstrap intervals quantify finite-seed uncertainty conditional on this model; quadrature, grid resolution, and model mismatch are separate uncertainties.

The measured operating limits show why a successful noiseless cancellation is insufficient for an experimentally accurate geometric phase. Both the biased energy integral and independent measurement fluctuations survive that cancellation. Assessing the complete reconstruction exposes substantially tighter limits in the interacting regime and a total sampling cost that is invisible in

endpoint depth. Measurement compilation, allocation, and control of integrated bias are therefore central requirements for turning adaptive circuit compression into a practical Berry-phase experiment.

Reproducibility. The accompanying directory contains the independent model and adaptive solver, circuit-noise kernels, finite-sampling and threshold scripts, all per-configuration/strength/seed records, reference cross-checks, validation results, and an explicit attempt log. Tables and figures are generated from these records. Table 4 lists the full strength inventory; a machine-readable form retains every seed and both phase branches.

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